

Softmax Regression

CGPA	IQ	Place	} multi class classification
7.1	71	0	
8.5	85	1	
9.5	95	2	

$$\sigma(z)_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\sigma(z)_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\sigma(z)_3 = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$0 < \sigma < 1$$

Loss Function

$$L = \frac{-1}{m} \sum_{i=1}^m y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)$$

m = no. of rows

Loss function for softmax

$$L = \frac{-1}{m} \sum_{i=1}^m \sum_{k=1}^K y_k^{(i)} \log(\hat{y}_k^{(i)})$$

x_1	x_2	y	$y_{k=1}$	$y_{k=2}$	$y_{k=3}$
x_{11}	x_{12}	1	1	0	0
x_{21}	x_{22}	2	0	1	0
x_{31}	x_{32}	3	0	0	1

$$\gamma_1^{(1)} \log(\hat{\gamma}_1^{(1)}) + \underline{\gamma_2^{(1)} \log(\hat{\gamma}_2^{(1)})} + \underline{\gamma_3^{(1)} \log(\hat{\gamma}_3^{(1)})}$$

$$+ \underline{\gamma_1^{(2)} \log(\hat{\gamma}_1^{(2)})} + \gamma_2^{(2)} \log(\hat{\gamma}_2^{(2)}) + \underline{\gamma_3^{(2)} \log(\hat{\gamma}_3^{(2)})}$$

$$+ \underline{\gamma_1^{(3)} \log(\hat{\gamma}_1^{(3)})} + \underline{\gamma_2^{(3)} \log(\hat{\gamma}_2^{(3)})} + \gamma_3^{(3)} \log(\hat{\gamma}_3^{(3)})$$

Underlined terms will become zero

∴

$$L = \gamma_1^{(1)} \log(\hat{\gamma}_1^{(1)}) + \gamma_2^{(2)} \log(\hat{\gamma}_2^{(2)}) + \gamma_3^{(3)} \log(\hat{\gamma}_3^{(3)})$$

$$\hat{\gamma}_1^{(1)} = \sigma(W_1^{(1)} x_{11} + W_2^{(1)} x_{12} + W_0^{(1)})$$

$$\hat{y}_2^{(2)} = \sigma (W_1^{(2)} x_{21} + W_2^{(2)} x_{22} + W_0^{(2)})$$

$$\hat{y}_3^{(3)} = \sigma (W_1^{(3)} x_{31} + W_2^{(3)} x_{32} + W_0^{(3)})$$

$$\text{Coeff} = \begin{bmatrix} W_1^{(1)} & W_2^{(1)} & W_0^{(1)} \\ W_1^{(2)} & W_2^{(2)} & W_0^{(2)} \\ W_1^{(3)} & W_2^{(3)} & W_0^{(3)} \end{bmatrix}$$

Gradient descent

$$W_1^{(1)} = W_1^{(1)} - \eta \frac{\partial L}{\partial W_1^{(1)}}$$

$$W_2^{(2)} = W_2^{(2)} - \eta \frac{\partial L}{\partial W_2^{(2)}}$$

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